Taylor & Francis
Taylor & Francis Group



Architectural Science Review

ISSN: 0003-8628 (Print) 1758-9622 (Online) Journal homepage: www.tandfonline.com/journals/tasr20

Mapping CAD education in architecture: a critical review and survey of bachelor programmes in Europe

Elena Vazquez

To cite this article: Elena Vazquez (11 Nov 2025): Mapping CAD education in architecture: a critical review and survey of bachelor programmes in Europe, Architectural Science Review, DOI: 10.1080/00038628.2025.2584178

To link to this article: https://doi.org/10.1080/00038628.2025.2584178

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 11 Nov 2025.

Submit your article to this journal

Article views: 156

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tasr20

ARCHITECTURAL SCIENCE REVIEW
https://doi.org/10.1080/00038628.2025.2584178

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Mapping CAD education in architecture: a critical review and survey of bachelor programmes in Europe

Elena Vazquez

Umeå School of Architecture, Umeå University, Umeå, Sweden

ABSTRACT

This study examines the current state of computer-aided design (CAD) in architectural education, revisiting early 2000s debates on digital design integration in Bachelor of Architecture programmes. Using a mixed-methods approach, the study analyzes curricula from 50 European architecture schools and reviews foundational scholarship on CAD education. The study examines how, when, and what aspects of digital design are taught, considering rapid advances in AI and the growing emphasis on sustainable design. Two main methods of CAD integration are identified: through dedicated courses and integrated into existing courses, with schools split between these approaches. Most dedicated digital design courses occur in the first years, suggesting a shift toward early integration. However, CAD remains largely framed as a representational tool since it is frequently embedded in drawing courses. Despite growing scholarly interest in early integration of advanced topics such as parametric design, current curricula primarily emphasize general digital design methodologies.

ARTICLE HISTORY
Received 18 June 2025
Accepted 20 October 2025

KEYWORDS
CAD education; Computer-Aided Design (CAD); architecture education; digital design pedagogy; basic design education; Architectural curriculum

Introduction

Computer-aided design (CAD) has been an integral part of architectural practice and education for the past few decades. CAD tools have steadily advanced within a constantly shifting technological landscape, with recent breakthroughs – particularly in AI – now driving applications that range from optimizing design processes to inspiring creative innovation (Del Campo and Leach 2022). In parallel with these advancements, CAD has increasingly permeated architecture curricula at universities, albeit to varying degrees. During the eCAADe (Education and Research in Computer-Aided Architectural Design in Europe) conferences of 2000, 2001, and 2002, a series of timely discussions took place about digital design in architecture curricula (Mark, Martens, and Oxman 2001). These discussions took place at a time when a third generation of CAD software was becoming available, characterized by advanced 3D modelling and the emergence of Building Information Modeling (BIM) (Caetano and Leitão 2020).

Since then, a limited number of studies have addressed the revision of architecture curricula to better integrate CAD. In 'The Ideal Computer Curriculum,' Mark, Martens, and Oxman (2001) propose a framework for integrating computer technology into a design education programme. They proposed two scenarios: one set within the framework of curricula at the time (the integrated approach) and the second, in which several traditional courses are displaced (the replacement approach). The new proposed digital courses were set at three levels: basic (e.g. digital design media), intermediate (e.g. geometric modelling and rendering, digital moviemaking), and advanced (e.g. performance simulation). In the integrated approach, the new courses are mostly integrated within the architectural design studio. In the replacement approach, the traditional architectural design studio is replaced by specialized classes at different levels.

In another study, Duarte, Celani, and Pupo (2012) describe two cases of integrating computational design into the undergraduate architecture curricula: one in UNICAMP (Universidade Estadual de Campinas) and another one in IST (Instituto Superior Técnico, Lisboa). In the UNICAMP case, CAD content was introduced through dedicated, specific topics (such as generative design and digital fabrication) – a dedicated CAD courses approach. In the IST case, CAD content was integrated with architecture content through two dedicated CAD courses, and additional CAD content was incorporated into existing courses – a hybrid approach. In both cases, however, computer applications are not seen as representational devices; instead, they are conceptual tools. The authors argue that the success of integrating computational design into the curricula depends mainly on the cultural and organizational aspects of institutions.

More recently, the COVID-19 pandemic accelerated the shift to digital and hybrid teaching environments, as educators were forced to adopt online teaching platforms (Geyer 2024). During these times, several digital collaboration tools were implemented, including Miro boards and other learning management systems (Mariotti and Niblock 2023). The transition to a virtual design studio may also have signified an increased literacy on design software among students, as reported in Iranmanesh and Onur (2021). Online environments and tools adopted in virtual studios allowed for the co-creation of knowledge (Pelsmakers et al. 2021), and many of these tools remain in use today. Nevertheless, aside from the growing adoption of collaborative digital

CONTACT Elena Vazquez Elena.vazquez@umu.se Umeå School of Architecture, Umeå University, Östra Strandgatan 30 C, 903 33, Umeå, Sweden

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

tools in teaching environments, there is limited discussion in terms of structural changes to CAD teaching.

Despite earlier efforts to integrate CAD into architectural education, recent years have seen limited attention to the current positioning of digital design within architecture curricula. An exception is Soliman, Taha, and El Sayad (2019), who studied the integration of CAD in the curricula of 20 architecture schools (ranked among the top 20 globally) and eight architecture departments in Egypt. Nevertheless, the sample size was limited, and the selected universities are spread out across the globe. Focusing on a region that shares a similar structure of higher education, such as European universities with the Bologna process, might paint a clearer picture of the state of the art. Additionally, reviewing the literature on digital design integration at the bachelor's level may provide a point of comparison between contemporary digital design pedagogy and the current status of CAD in educating architects in Europe.

While the importance of CAD in architecture education may seem self-evident, it continues to play a central role in the curricula for several key reasons. First, CAD is a required competence in the labour market (Salleh et al. 2013). Second, a growing body of architectural concepts has emerged whose theoretical foundation is digital design, including generative and performative processes that are not available in paper-based methods (Oxman 2008). Finally, there is very little architectural work today that does not use a computer in some way – whether for representation or as part of the design process.

However, no research has been found that surveys the current state of CAD integration in the architecture curricula, while critically revisiting the discussion of CAD integration in architecture education. The relevance of revisiting CAD in architectural curricula becomes even more pressing today, as we enter a new phase marked by accelerated AI integration. Recent advancements – such as text-to-image generation, AI-enhanced design software, and automated floorplan generation (E.g. PlanFinder 2025) – are reshaping the possibilities of architectural design. Despite the current skepticism surrounding digital technologies, including computational design (Carpo 2024), AI is being increasingly integrated into everyday CAD and graphic design software, enabling new design workflows to emerge and transform architectural practice. In addition, there is an increasing interest in bio and natural materials in architectural research in the current climate crisis (Chayaamor-Heil et al. 2023), which suggests that a better understanding of digital tools may help guide the next generation of architects in efficient and creative material resource use.

Almost 25 years after the eCAADe roundtable discussions on CAD in schools of Architecture, this article presents a reflection on the formal integration of digital design in the Bachelor of Architecture curricula in the European context. This research has two main objectives: to examine the current integration of digital design within the Bachelor of Architecture curricula (based on an analysis of 50 architecture schools across Europe) and to revisit the discourse on digital design in architecture education. The study investigates three key aspects of digital design integration in Bachelor of Architecture curricula in Europe: How is digital design incorporated? (Research Question – RQ1), When is digital design introduced? (RQ2) What are the CAD topics being taught across the different years? (RQ3). These three questions are addressed through a survey and a discussion that compares the findings with contemporary discourse on the subject.

This article is focused on CAD within the context of Architecture education, specifically, the Bachelor of Architecture education – the foundational three-year programme offered by schools of Architecture. Digital design integration in master's programmes has progressed rapidly – often serving as a testing ground for novel pedagogical approaches (Oxman 2008). Examples of recent scholarship advancing pedagogical approaches for integrating digital design at the graduate level can be found in (Balbo 2024; Jenny et al. 2022). In contrast, digital design integration at the bachelor's level is relatively underexplored, following perhaps more traditional pedagogical models. This article is therefore focused on the bachelor's level, considering its foundational role in shaping generations of architects, providing the basic building blocks of architectural training. In this study, the term CAD – used interchangeably with digital design throughout the article – is understood broadly as a foundational skill for architecture students, rather than a niche specialization. Following Doyle and Senske's (2017a) definition of digital design, CAD is understood as the use of computers and computer-controlled tools in architectural design.

Materials and methods

This paper adopts a mixed-methods approach, combining data collection and analysis from architecture programmes in Europe with a comprehensive literature review in the discussion of the results. The literature review was conducted by searching databases using keywords and Boolean operators such as: 'Digital Design' and 'Architecture education'; 'CAD' and 'Architecture pedagogy'; 'Computational Design' and 'Architecture education'. The databases used were Google Scholar and CumInCAD (publications in Computer-Aided Architectural Design by eCAADe and its affiliated associations). The literature selected was focused on Bachelor of Architecture Education. Interior design and landscape architecture were also excluded from the search, as well as civil engineering and related fields that might also incorporate CAD education in their higher education programmes.

The data collection from architecture programmes was conducted in the spring of 2025. It was limited to the 50 best-ranked architecture schools, according to the QS World University Rankings for Architecture and the Built Environment, with a filter on region: Europe. The first 50 schools in the ranking above are located in the UK (15), Italy (6), Germany (6), Spain (4), Sweden (3), Netherlands (2), Switzerland (2), Belgium (2), Austria (2), Portugal (2), Denmark (2), Norway (1), Greece (1), Ireland (1), Finland (1). Of these schools, the majority (66%) are members of the EAAE-European Association for Architectural Education, which currently has around 140 member schools.

To answer RQ1 (How is digital design incorporated?), the course titles from the different programmes were analyzed, searching for dedicated digital design courses. In addition, all the course content was examined, mapping the occurrence of key terms in computer-aided design, and pairing the keywords to the courses in which they were integrated. To answer RQ2 (When is digital design introduced?), data was collected in terms

Table 1. Keywords extracted and terms included in them.

Keywords	Includes
Digital methods	Digitalization, digital tools/approaches, information technology, computer techniques/digital skills/computational tools
Digital representations	Digital communication, digital representation, digital media, digital production, digital drawing, Computer-aided drawing, software-based representation, computer graphics, digital sketches
3D model	Digital model, computer model, 3D CAD, 3D tools, 3D representations
Digital Fabrication	3D printing, digital prototyping
CAAD	CAD, 2D CAD, Computer-Aided Design, Computer-aided Architectural design.
Programming	Programming, algorithms
Building Performance Simulation (BPS)	Optimization, computer simulation, simulations, energy simulations, energy models.

of what year(s) the dedicated courses are placed within the curricula. To answer RQ3 (What are the CAD topics being taught across the different years?), CAD keywords were extracted from programme documents, pairing the keywords with the corresponding year they came from.

The curricula for the Bachelor of Architecture Programs were analyzed by examining both the course titles and course documents. The data was collected by analyzing course syllabi extracted from course catalogues and programme websites. The language of instruction of the programmes varies by country; therefore, machine translation was used to translate some programme documents. The search was limited to the first 3 years of every programme. The availability and depth of course information varied across institutions. A small portion (4%) provided only brief course titles with minimal descriptions. About 32% offered extended course descriptions, averaging 130 words, while 24% included both detailed descriptions and learning outcomes, with an average of 365 words. The remaining 40% presented comprehensive course information, including descriptions, learning outcomes, examination methods, and other relevant details, averaging 412 words per syllabus. Overall, most institutions made course titles and extended descriptions accessible online, and all programme websites included at least a short course description. Keywords such as 'computing,' 'digital,' 'computers,' and 'CAD' were implemented as search queries among course titles, course syllabi, and course documents. The goal was to assess and measure explicit integration in the curricula only. Digital design could be, however, integrated in courses beyond those explicitly mentioned in course titles and publicly available course descriptions and programme documents.

To identify and quantify the occurrence of the different topics of digital design, keywords were extracted from the course titles and the course documents. Only keywords related to CAD were extracted, as shown in the table below. After the keywords were extracted from the course content and description, similar keywords were grouped; for instance, 'digital communication', 'digital drawings', and 'computer-aided drawings' were included under the umbrella of 'Digital representations' (Table 1).

Results

The results below are grouped into How, When, and What. The first section (How) discusses how digital design is incorporated in the different schools surveyed. The second section (When) documents the state of the art regarding when digital design is introduced into the curricula in the surveyed schools. The third and final section (What) presents a thematic analysis of digital design in the surveyed curricula.

How

Figure 1 presents the distribution of strategies employed by architectural programmes to incorporate digital design into their educational programmes. Half of institutions integrate digital design within existing courses (50%), suggesting a preference for embedding digital competencies into existing pedagogical structures. These institutions do not offer dedicated CAD courses; instead, they embed CAD content into other classes. Meanwhile, 28% of programmes adopt a more focused approach through dedicated courses, and 20% implement a hybrid model that combines both dedicated and integrated courses. Only 2% of the surveyed curricula report no explicit integration of digital design; however, this does not necessarily mean that there is no mention of CAD or digital design in the public documents describing the programme. In programmes where dedicated CAD courses exist – 48% of schools have dedicated and hybrid approaches – the non-elective digital design courses amount to an average of 6.75 ECTS (European Credit Transfer and Accumulation System). Note that most bachelor's degrees equal 180 ECTS.

In terms of what courses CAD is integrated into, Figure 2 illustrates the thematic distribution of digital design integration within non-dedicated courses in architectural curricula. The data reveals that drawing courses constitute the primary context for digital design integration, accounting for 37% of the instances. This includes courses titled (architectural) representation, communication, visualization, and so on. This is followed by design studios (23%) and basic design (11%), indicating a significant role in embedding digital tools within core design activities. Form/geometry courses each account for 14%, suggesting some preference for CAD integration in foundational and formal explorations. Other thematic areas – design skills, art and design, urban studies ('City'), and environmental sciences, as well as architectural technology – each comprise 3%, reflecting a more limited but diverse application of digital design across the curricula. These findings highlight the predominance of representational contexts in the integration of digital design.

When

Digital design integration in the curricula is made through either dedicated courses or integrated into the more traditional architectural education coursework (drawing, studio, and so on). The schools that provide dedicated coursework in digital design

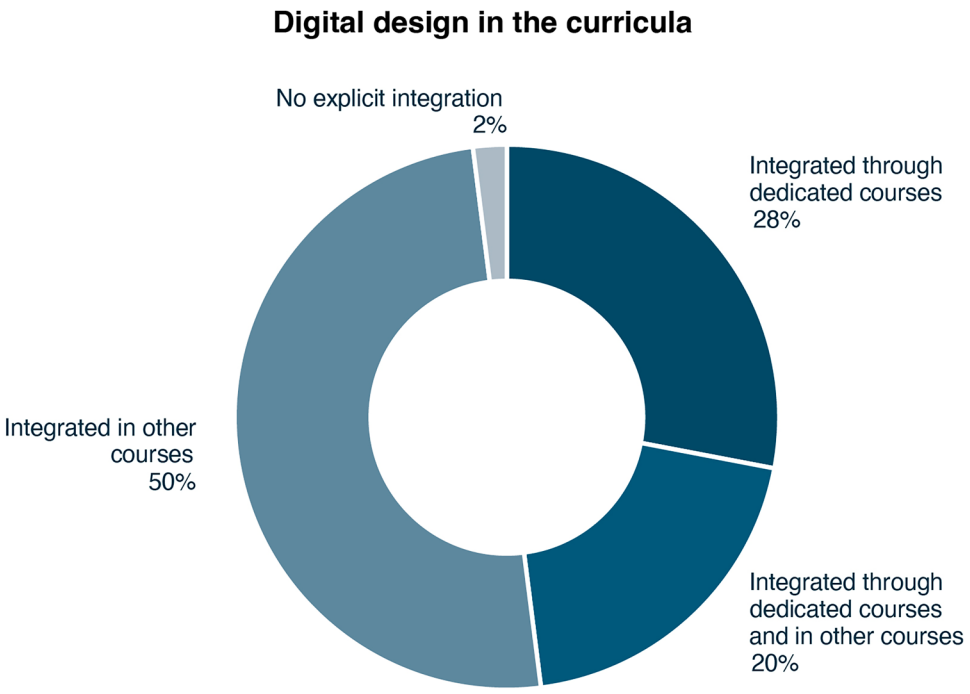


Figure 1. Digital design integration strategy. Percentages are based on 50 surveyed schools.

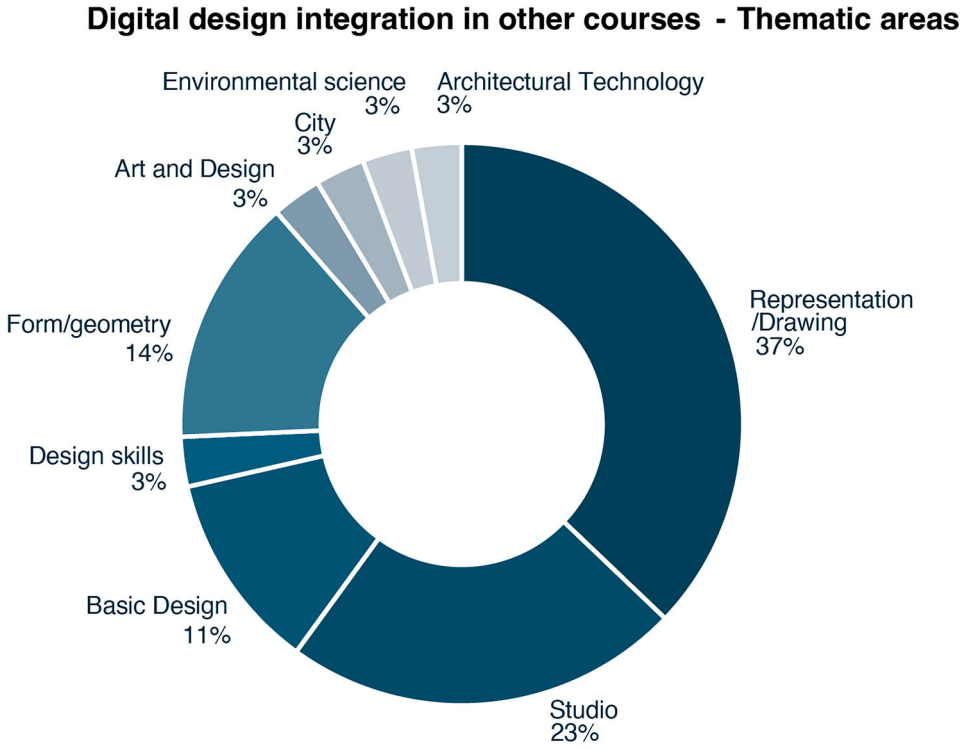


Figure 2. Digital design integration in existing courses. Data reflect 70% of the sample (35 schools), including 50% with CAD integrated in other courses and 20% using a hybrid approach.

account for about half of the surveyed schools. These are schools that offer CAD courses as part of their curricula (mandatory courses). From these schools that offer dedicated courses, most of the courses on digital design are placed in the context of the first years of study in Architecture, as shown in Figure 3. Most programmes place dedicated CAD courses in the first and second year (32%). Another trend is to place CAD courses either only in the first year (18%) or only during the second year (27%). Overall, the results suggest that a minority of dedicated CAD courses are offered during the third year of study. Note that these exclude elective CAD courses, which were found to be placed in the third year of studies within the curricula.

Regarding courses with CAD integration (courses not dedicated to CAD, e.g. architectural drawing and studio), the

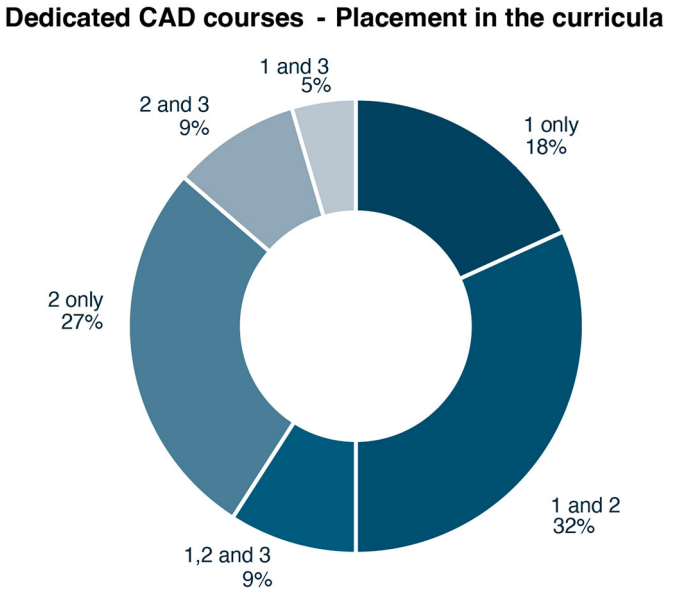


Figure 3. Survey results, dedicated CAD courses placement in the curricula.

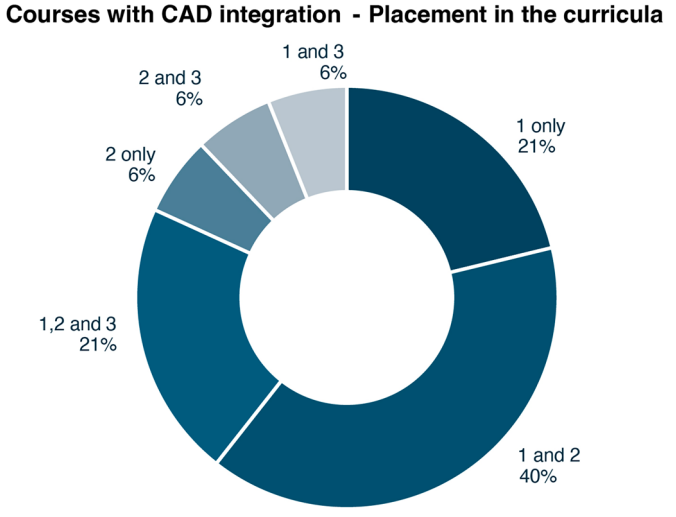


Figure 4. Survey results, courses with CAD integration, placement in the curricula.

placement of such courses follows a different trend, as shown in Figure 4. These are subjects in which the course documents – description and/or learning outcomes – refer to some area of CAD learning. Most courses with explicit CAD integration are placed both in the first and second year (40%), followed by courses with CAD integration placed in all years (21%), and in the first year only (21%).

What

CAD tools can be loosely grouped into those related to design (e.g. 3D modelling, parametric design, Building Information Modeling or BIM), fabrication (e.g. robotics, 3D printing), evaluation (e.g. Building Performance Simulation or BPS), and representation (e.g. AI, animation, and film). Fabrication is included as a subtopic of digital design in the survey because it is closely intertwined with CAD and is commonly used in digital design workflows. Note also that some digital design tools correspond to more than one category, such as BIM, which is used for representation, evaluation, and design. CAD keywords from these different subtopics were identified in course documents across the schools surveyed.

In terms of keywords in the titles of dedicated CAD courses within the curricula surveyed, the bar chart in Figure 5 illustrates the distribution and recurrence of the different keywords. Recall that 50% of the programmes surveyed had dedicated digital design courses. One of the most frequently occurring titles is 'Digital Methods/tools,' appearing in eleven instances, which naturally allows for the inclusion of a broad range of subtopics under the umbrella of digital methods. Also with an equal number of appearances is 'CAD + Digital design', which also suggests courses in foundational CAD. 'Digital representation' appears six times, indicating a strong emphasis on CAD as a representational tool, and '3D modelling' is also a recurrent subject. Other course titles, such as 'Generative Design,' 'Parametric Design,' and 'BIM,' are less common. This distribution suggests a curricular focus on general digital methodologies, with more specialized topics receiving comparatively limited attention.

Figure 6 shows the distribution and recurrence of CAD keywords within elective digital design course titles. The diversity of titles reflects a broad engagement with emerging technologies and computational methods, rather than foundational topics, which was the case with the dedicated courses described above. Digital design and digital fabrication are the most recurrent keywords in the titles of courses offered as electives. In addition, 'Parametric design' is a recurring elective course provided, followed by 'Building Performance Simulation' and 'Programming'. If we merge the topics 'Parametric Design' and 'Programming', this would be a large group, showing a trend towards educating architects on the fundamentals of programming for design, whether as textual code or through VPLs (Virtual Programming Languages). Less common are topics such as 'CAD Geometry,' which appears rather foundational, as well as more specialized subjects like 'Robotic Fabrication' and 'AI.'

The chart in Figure 7 illustrates the distribution of digital design-related keywords across three academic years (Y1, Y2, Y3) within courses where digital content is integrated rather than taught in standalone classes. These keywords were collected from the course content, rather than course titles. Recall that half (50%) of the schools integrate digital design into other courses. 'Digital methods' and 'Digital representations' are the most frequently referenced terms, particularly in the first and second years, indicating a foundational emphasis on general digital literacy and visual communication. '3D model' and 'CAD' also appear prominently, reflecting their relevance in early design education. More specialized terms, such as 'Parametric' and 'BIM/Digital twin,' show increased presence in later years, suggesting a progressive integration of advanced tools. Surprisingly, 'Render' appears as frequently as 'BIM', which might indicate a standing disregard towards BIM tools in architecture schools. 'BPS' tools are still relatively uncommon but increasingly appear in later years. Other emerging technologies, such as 'AI,' 'AR (Augmented Reality),' and 'Robotics,' are mentioned less frequently, indicating limited curricular experimentation.

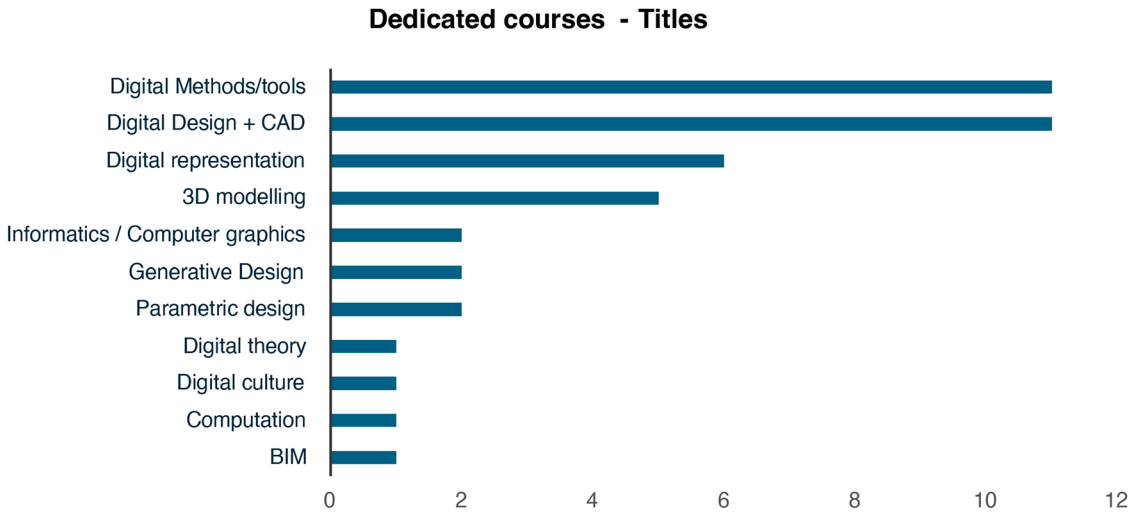


Figure 5. Dedicated courses in digital design: titles. Bar chart values represent the number of occurrences.

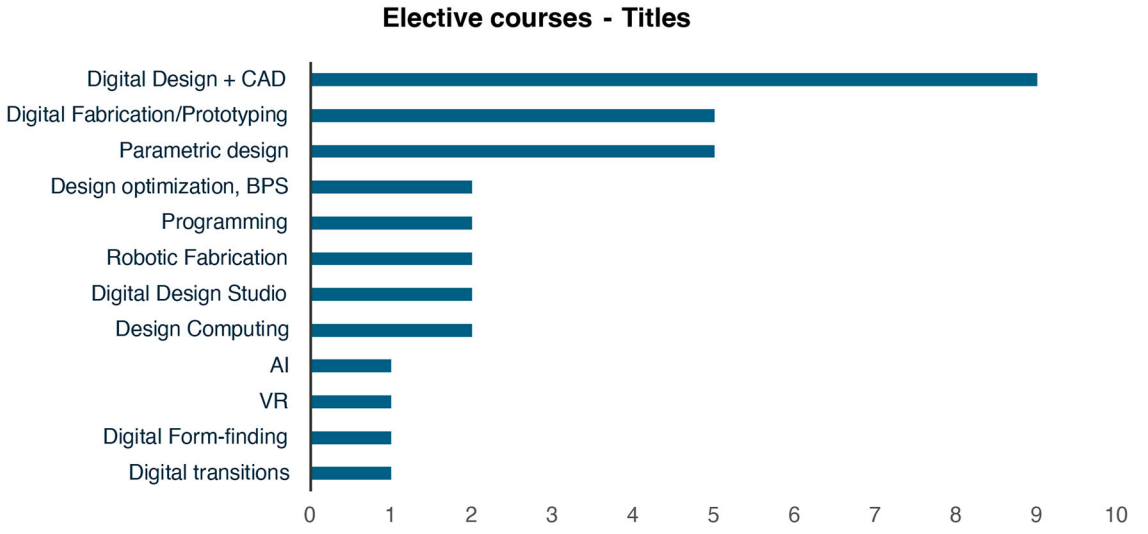


Figure 6. Elective courses, titles. Bar chart values represent the number of occurrences.

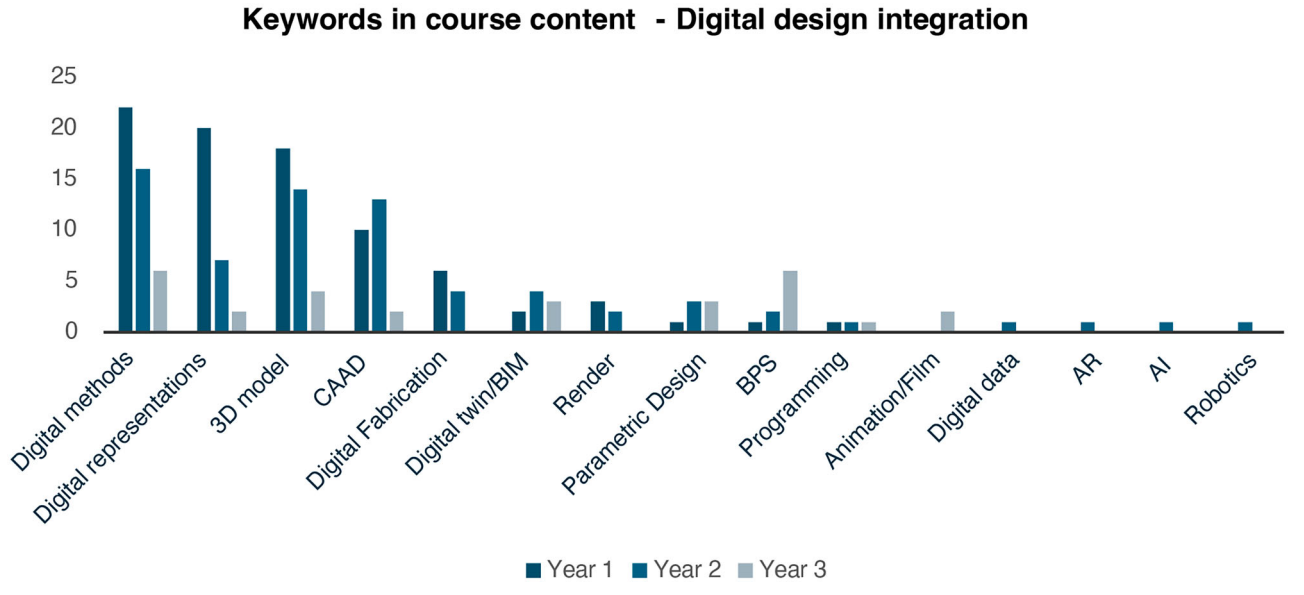


Figure 7. Keywords in course content across the three years of the BA Architecture programmes.

Discussion

How

There are two general approaches to teaching digital design in architecture schools, according to (Marx 2000): as a standalone course (the dedicated course approach) or integrated into the studio (the integrated approach). There is, however, a third option of CAD being taught both through dedicated courses and integrated into studio or other classes (the hybrid approach).

The dedicated course approach entails teaching CAD through standalone courses. Marx (2000) argues that the best approach is to have digital subjects in standalone classes, due to the limited time students have within the design studio, the lack of digital design skills among faculty members, and the high cost of providing computers for students in the studio space. Another reason to advocate for a dedicated course approach lies in the range of knowledge types required to learn CAD effectively. As outlined by Chester (2008), these include declarative knowledge (e.g. commands), procedural knowledge (e.g. how to create a cylinder), and strategic knowledge (e.g. metacognitive processes such as planning and revising). Traditional learning methods typically emphasize procedural and declarative knowledge. Nevertheless, it is essential to teach students how to transfer procedural knowledge to other software and focus on the strategic understanding of applying their CAD knowledge for their own design goals. While all these types of knowledge can be addressed in both dedicated courses and design studios, integrating them within design studios or other courses may pose challenges due to time constraints, which could potentially limit the depth and breadth of coverage.

Another advantage of the dedicated coursework approach is that it would reduce the dependence on self-learning, which is currently reported in architectural practice in Europe (The Architects' Council of Europe 2025). Not offering structured training in digital design may also perpetuate existing gender inequalities in digital technologies in architecture, which have been discussed in (Doyle and Senske 2017b; Doyle, Forehand, and Senske 2017). Increasing digital literacy through structured CAD teaching can help democratize access and address the gender gap in technology.

The integrated approach is when CAD teaching is embedded into existing design studios or other courses, such as drawing courses. Kara (2015) argues that digital design integration is better done through a studio, where the knowledge becomes more permanent through application in design projects. It is indeed indispensable to adopt a project-based approach to teaching CAD, in line with contemporary pedagogies that favour student-centered approaches. Several accounts exist of integrating various digital tools in architecture studios, including digital prototyping (Petric and Maver 2003), Virtual Reality (VR) (Oprean et al. 2018), and parametric design (Schnabel 2006). The design studio can be a fertile ground for exploring different emerging tools, without having to commit to changes in the syllabus or curriculum, which is the advantage of this approach to integration. Nevertheless, a drawback of this approach is that CAD integration in architectural design studios may require tutors to understand the potential of computing in design and a willingness to guide students in this direction (Mark, Martens, and Oxman 2003).

The survey reveals that schools are primarily divided in their approach to CAD teaching, with about half of the schools surveyed offering dedicated CAD courses as part of their core curricula, including 28% teaching CAD only through dedicated courses, and 20% integrating CAD both in other courses and through dedicated courses (the hybrid approach). Overall, CAD content is found to be explicitly part of the architecture education curricula in almost all schools surveyed. Nevertheless, the survey results highlight that there is a prevalent view of digital design primarily as a representational tool, considering that most (37%) of schools that integrate CAD into other subjects incorporate it in drawing courses. Scholars in computational design theory have long proposed integrating CAD software beyond its representational function (Duarte, Celani, and Pupo 2012; Mitchell 1975). Architectural education, however, at least at the foundational level, has yet to fully embrace the idea that digital design is both conceptually and practically instrumentalized. The reason why there is still a focus on the role of digital design in architectural representation may be that older paradigms of design are strongly reliant on visual reasoning, conceiving design as a reflection supported by (paper-based) representational processes (Oxman 2008). Digital design offers a different medium, capable of intertwining processes of design, fabrication, and evaluation.

In summary, some scholars advocate for dedicated courses due to logistical constraints and the need for focused skill development. In contrast, others emphasize the pedagogical benefits of embedding digital design within studio environments to foster deeper, project-based learning. The literature highlights the studios' potential as a flexible space for experimenting with emerging technologies, while also noting the challenges posed by faculty readiness and time limitations. Nevertheless, more important than whether it is integrated into studios or dedicated courses is that CAD teaching adopts a project-based approach, ensuring that learning extends beyond skill building into strategic CAD thinking. Indeed, Mark, Martens, and Oxman (2003) argue that CAAD teaching should move beyond skill building, an approach that broadens the separation between CAAD as a tool and architecture. A project-based approach must therefore be applied, following contemporary pedagogical approaches as discussed in (Chester 2008; Oxman 2008; Vazquez 2024).

When

In terms of when to integrate CAD into architecture curricula, CAD has traditionally been excluded from foundational design courses, a stance that has been reconsidered in light of its evolving role in architectural practice. CAD software, initially conceived as (2D) drafting tools, enables the creation of holistic three-dimensional models today instead of reducing architecture to orthographic projections. In contemporary architectural practice, digital design is often used in a functional way for early form exploration, integrating performative aspects such as structural and environmental performance (Abdullah and

Hassanpour 2021). Scholars have presented different views on when CAD should be incorporated into Architecture education and its relation to basic design education.

Kara (2015) argues that hand drawing and physical model making should be prevalent in the foundational levels, and digital tools should be introduced after these are established. Through a comparison of three schools – two in the USA and one in Turkey – the author makes a case for integrating digital design from the second year onwards in architectural education. The reason to do so, the author argues, is so that students acquire spatial and tectonic knowledge that comes from hands-on making. Nonetheless, this position assumes a false dichotomy: digital design need not preclude hands-on making. Furthermore, the author claims that there is a delay in visualizing the results in digital design, which hinders creativity in some way that manual making does not due to the immediacy of the results. Yet, experienced users of 3D modelling software are aware of its capacity for rapid iteration and real-time visualization. There seems to be no reason for a strong foundation in hand drawings not to be cultivated in tandem with digital literacy, allowing for shifts between mediums.

The literature on CAD and creativity also identifies potential drawbacks associated with CAD use, which may explain why some scholars advocate for delaying the introduction of CAD in architecture education. Robertson, Walther, and Radcliffe (2007) identify several unintended consequences of CAD use on student creativity. These include circumscribed thinking, where software constraints limit design possibilities; premature fixation, in which early detailing leads to attachment to specific solutions; and bounded ideation, where prolonged CAD use under pressure diminishes motivation. Another drawback is related to how dealing with CAD menus and toolbars hinders the generation of creative ideas by shifting the focus from the task to the tool (Okeil 2010). However, this effect appears most pronounced during the initial learning phase. As proficiency develops, the cognitive focus tends to return to the design task, with the tool becoming more transparent in the creative process.

The introduction of CAD at foundational levels also leads to a discussion of its relation to basic design education in the curriculum. Uysal and Topaloğlu (2017) describe how, in most curricula, digital design education is detached from basic design education, resulting in CAD becoming a tool solely for representation and production, not for design. If detached from basic design education, the authors argue, students find it hard to connect the sensitivities they developed through basic design projects with the competencies acquired through engagement with CAD. Similarly, Duarte, Celani, and Pupo (2012) argue for learning hand drawing skills in parallel with digital design, so that they become proficient in both and use them intertwined in the design process. They suggest integrating CAD subjects as early as possible. To help translate the material and aesthetic sensibilities acquired during basic design education, students would appear to benefit from a better integration of basic design and CAD education.

Manual and digital tools are sometimes framed in opposition, yet they prompt different dimensions of creativity (Pressman 2012), and thus should be seen as complementary rather than conflicting. The subject of CAD use and creativity is indeed a well-studied area of inquiry; Botchway, Amos-Abanyie, and Afram (2015) argue, for instance, that early design explorations can be creatively enhanced with the use of CAD tools. In addition, Ozkar (2017) states that computation in architecture is now a central part of discussions on creative reasoning and a key contemporary issue in design pedagogy. Creativity and CAD use have also been studied using psychometric approaches: For instance, Dawoud, Al-Samarraie, and Zaqout (2015) studied the relationship of flow – a motivated and self-determined desirable state that can occur during the design process –, creative behaviour, and CAD use. The researchers concluded that CAD can trigger a state of deep engagement (flow), which in turn can predict creative behaviour. Considering how CAD can facilitate creative thinking and behaviour, it seems clear that it should be an integral part of foundational design education.

An approach that concurrently integrates basic design education and digital literacy has already been implemented in many schools across Europe: over sixty percent of dedicated digital design courses take place during the first year of studies. In addition, when looking at the courses with CAD integration and their placement in the curricula, the survey reveals that most schools integrate CAD content both in the first and second year (40%), in the first year (21%), and throughout the three years of the programme (21%). The early integration of CAD into architectural education signals a shift toward aligning pedagogy with contemporary practice and the competencies it demands.

The sequential model of introducing manual tools followed by digital tools may not align with contemporary professional workflows, nor take advantage of cognitive processes associated with creativity that can be cultivated and enhanced through the use of CAD. The concurrent development of basic design education and digital competencies may better prepare students for the multifaceted demands of contemporary architectural practice, as reflected in the survey results.

What

Digital design in architecture encompasses a wide array of subtopics and tools, a diversity clearly reflected in current trends within professional practice. In *The Architectural Profession in Europe* report (The Architects' Council of Europe 2025), we can see that 3D modelling tools remain the most popular tool, used by about half of architects, and that over a quarter regularly use BIM. Other commonly used tools are rendering tools, building performance simulation tools, and parametric design tools. Scholars have not equally examined the range of digital design tools used in practice in architectural pedagogy. Additionally, these tools are not uniformly integrated into the various architecture programmes surveyed, as discussed below.

A recurrent theme of discussion in digital design pedagogy is whether all architecture students should acquire basic programming skills. Celani (2008) discussed the relevance of incorporating computer programming in the architecture curricula, arguing that programming can improve logical reasoning and conceptual thinking in design. In addition, the author argues that programming can be used as an administrative tool to develop implementations, integrating knowledge from different courses. Leitão, Cabecinhas, and Martins (2010) also argue for incorporating programming into the architectural curricula to overcome the limitations of current CAD software. Visual programming

languages are typically more popular among architects and are commonly associated with parametric design. For instance, Ostrowska-Wawryniuk, Strzała, and Słyk (2022) describe a standalone geometry course using parametric design for second-year architecture students (Digital Descriptive Geometry) at Warsaw University of Technology. The course emphasizes algorithmic thinking and problem-solving, and the authors argue that this method improved students' understanding of computational design processes.

BIM is a subtopic in digital design that relates to both design and evaluation, combining three-dimensional models with a reference database of materials and construction information. BIM software has become the industry standard in many regions of the world, which is why it is inherently professionally oriented. In general, there is limited discussion on the integration of BIM in architecture education in the foundational years. Regarding BIM in the broader Architecture, Engineering, and Construction (AEC) field, Abdirad and Dossick (2016) conducted a systematic review of the literature on BIM integration in AEC education, identifying pedagogical strategies for it. BIM can also be connected to VPL and parametric design software. For instance, (Mikhailov et al. 2020) propose a method for implementing BIM and parametric modelling in a second-year architectural design course.

In the realm of digital fabrication in architectural education, several machines have emerged in architecture schools worldwide over the past few decades, including 3D printers, numerically controlled cutters, and robotic arms, reflecting an increasing interest in digital fabrication. In 'Beyond Digital' (Carpo 2023), Carpo argues that computational design would have to tackle architectural construction to ease what he describes as the fabrication gap between the highly sophisticated digital design tools and fabrication tools available. He frames this shift as essential, driven by the urgency of the climate crisis and the challenges of housing affordability. Digital fabrication has enabled a greater integration of design and fabrication, where material logic can be incorporated into the genesis of design processes. Oxman and Oxman (2010) argue that we should educate architects to function as material practitioners, understanding fabrication techniques as a vessel for creative exploration based on material constraints.

In the subtopic of evaluation tools in digital design, Building Performance Simulation (BPS) has gained increasing attention in the current context of the climate emergency. In terms of BPS pedagogies, J Hopfe et al. (2017) conducted a survey to understand how BPS was taught at universities. The findings indicate that most educators are concerned with daylight and thermal performance, as well as airflow. Structures are underrepresented, which is probably related to the ease of use of available software. The authors noted that, at the time, BPS tools were not user-friendly. Recently, however, software companies have placed great emphasis on BPS for early design and ease of use, as seen in products such as Autodesk Forma and Climate Studio (Autodesk 2025; Solemma 2025).

Virtual reality (VR) has been utilized as a tool to guide the design process, providing immersive, interactive, computer-generated environments. It can also serve as a communication tool, facilitating engagement with stakeholders in the building industry. While VR research has emerged in recent years, it has not been formally incorporated into curricula until now, except for smaller experiments in the design studio, such as the one described in (Alatta and Freewan 2017; Oprean et al. 2021). Other examples of VR integration in architecture include the work by Özgen, Afacan, and Sürer (2021), who conducted an experimental study with 20 first-year interior-architecture students, finding that VR enhances problem-solving in interior architecture and thus should be considered a promising tool for basic design education.

With the recent advancements in AI, current debates around AI and architectural design centre on how or whether to formally incorporate it into the curricula of architecture schools. A recent study by El Moussaoui and Krois (2023) described a survey among students, educators, and practicing architects to assess attitudes toward the role of AI in architecture. Most respondents agreed that integrating AI in architecture education is crucial. Educators have increasingly experimented with incorporating AI in architecture studios. As an example, Kavakoglu et al. (2022) conducted a pedagogical experiment integrating AI in early design education, using AI to create new façade representations based on a dataset generated by students' own work. Another study describes the use of AI rendering tools for applying case studies to 3D models in an architectural design studio (Alymani 2024). Regarding AI integration in curricula, Calixto and Croffi (2024) argue for a layered approach pairing computational design thinking and AI-driven education at similar abstraction levels. The author also contends that ideally, students should gain proficiency in visual programming to control and develop their own computational design and AI tools, rather than relying on third-party solutions.

Considering the recent emergence of AI, it is crucial to question what role we want future architects to have when engaging with these tools. If students were to become tool makers rather than simply tool users, they would benefit from having a basic understanding of programming. Being toolmakers would involve, for instance, composing their own datasets and then utilizing AI for design generation and optimization. Having at least the opportunity to learn basic programming as an elective course or workshop would increase computational literacy among architecture students and the number of tool makers in the next generation.

The survey results reveal that a broad spectrum of digital design topics is currently taught at architecture schools. Scholars advocate for integrating programming and algorithmic thinking to enhance design logic and toolmaking; nevertheless, these remain mostly absent from the core topics in the surveyed curricula, appearing instead as elective topics. BIM is recognized for its professional relevance, but it is hardly integrated into architectural education, as indicated by the survey results. The lack of BIM education represents one of the most significant gaps between architectural curricula and professional practice. Digital fabrication is framed as a means to foster material-based creativity among scholars, and it is a recurrent topic integrated into other courses and the subject of several elective and dedicated courses. The presence of digital fabrication in architectural curricula is likely due to its recognized pedagogical value and its role as a creative design tool. BPS and VR are increasingly seen as valuable for performance evaluation and immersive design exploration. AI has a limited presence in the surveyed curricula, primarily in elective courses and integrated into course content.

Table 2. Digital design courses, classifying them into digital tools related to design (D), fabrication (F), visualization (V), evaluation and analysis (E), and Surveying (S). Proposed learning outcomes are shown in the lower part of the table.

D	F	V	E	S		
■		■			Digital Design Media	Extracted from Mark, Martens, and Oxman 2001
		■	■		Geometric Modelling and rendering	
■		■			Digital Media and Drawing	
			■		Structural Analysis	
■		■			Digital Moviemaking and Animation	
			■		Computables of Design	
	■				Spatial Simulation Techniques	
■					Computer aided manufacturing and robotics	
				■	Digitalization of the third dimension	
			■		Laser surveying and photogrammetry	
			■		Performance simulation energy	
			■		Performance simulation acoustics and synthesis	
			■		Performance simulation: artificial and Daylight Representation	
■		■			Digital Technology and Communications Media	
	■				Computation of Construction	
				■	Geographic Information Systems	
			■		Spatial and Data Analysis Methods	
■		■			CAD I: Geometry and Visualization	Extracted from Duarte, Celani, and Pupo 2012
	■	■			CAD II: Programming and Digital Fabrication	
	■	■	■		CAAD: Computer Aided Architectural Design Studio	
■		■			Introduction to IT and Computer Graphics	
■		■			CAD Drafting and Modeling	
■		■	■		CAD in the Creative Process	
■		■			Modeling and Animation	
■			■		Integrated Collaborative Design BIM	
					Learners should demonstrate the ability to …	**Proposed learning outcomes**
■					Create 3D models of built environments with various geometries and modify their properties using modelling tools.	
	■	■			Generate building documentation, orthographic drawings, and visualizations from 3D models to both effectively communicate design intent and support the transition from design to construction.	
	■				Prototype designs, selecting adequate digital fabrication strategies (e.g. 3D printing and CNC).	
■	■		■		Create 3D data models of buildings using BIM and show familiarity with collaborative methods using BIM and cloud-based platforms.	
■			■		Select adequate building performance simulation (BPS) tools and use them to help guide design solutions.	
■					Implement computational tools in architectural design processes (e.g. parametric design, generative tools).	
				■	Obtain geographic data from a site and show familiarity with digital surveying methods for existing build environments.	
■		■	■		Select and critically incorporate AI tools in workflows for design, evaluation, fabrication, and visualization.	

However, the presence of AI as an elective topic may increase in architecture curricula worldwide in the future, considering the current accelerated development. Recent discourse also highlights the growing importance of AI in design education, with calls for its structured integration to prepare students for emerging technological paradigms.

The issue of what areas of digital design should be included in the curriculum is a critical one, as the process of updating or revising architecture curricula can be lengthy and complex. In contrast, the evolution of emerging technologies can seem immensely accelerated. Digital design education should encompass a range of subtopics, including design, evaluation, fabrication, and representation, at a foundational level. These competencies can be described as learning outcomes across the architecture bachelor's programme, regardless of whether they are implemented as standalone courses or integrated into other classes. Table 2 outlines a series of learning outcomes that encompass the described areas, including foundational aspects of digital design that can be explored sequentially throughout the Bachelor of Architecture programme. The table also compares the learning outcomes to courses in digital design as proposed by other authors. Other elective courses and shorter workshops can be used to address specialized knowledge such as AI for rendering and visualization, generative AI implementations, VR/AR, and programming basics.

Conclusion

This paper aimed to examine the integration of digital design in Bachelor of Architecture curricula. The survey of European schools found that half of them follow a embedded course approach, integrating CAD into other courses in their curricula, and the remaining half have dedicated CAD classes (28%) or offer a hybrid approach (20%). Structured CAD training may provide advantages in democratizing access to digital tools, but its effectiveness depends on adopting project-based, student-centered pedagogies. A notable trend is the incorporation of CAD into traditional drawing courses (37%), which contrasts with

the current discourse in computational design, which favours digital tools as operative rather than representational. The survey also indicates a tendency towards early integration, in line with contemporary digital design scholarship and current trends in practice. Regarding CAD content, the results suggest a predominant focus on general digital methodologies, with more specialized subjects, such as AI and parametric design, receiving comparatively limited attention.

The state-of-the-art survey was limited to publicly available information on architecture websites. The survey only measures explicit and formal integration of digital design in the architecture education of 50 schools in Europe, as reflected in documents such as syllabi and course descriptions. The survey, in this sense, does not capture CAD integration that occurs beyond these course descriptions, such as workshops not described in the curricula, or initiatives by faculty members to centre their pedagogy around CAD. Nevertheless, the data collected helps paint a picture of digital design education in architecture. Through the survey analysis and literature review, this article provides an overview of CAD education in architecture, identifying trends in pedagogical approaches at the bachelor's level. In general, the findings underscore the importance of developing curricula that not only adapt to rapid technological advancements but also foster critical and creative use of digital tools within architectural education.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available from the corresponding author, [author initials], upon reasonable request.

ORCID

Elena Vazquez http://orcid.org/0000-0003-1765-1259

References

Abdirad, H., and C. S. Dossick. 2016. "BIM Curriculum Design in Architecture, Engineering, and Construction Education: A Systematic Review." *Journal of Information Technology in Construction (ITcon)* 21 (17): 250–271.

Abdullah, H. K., and B. Hassanpour. 2021. "Digital Design Implications: A Comparative Study of Architecture Education Curriculum and Practices in Leading Architecture Firms." *International Journal of Technology and Design Education* 31 (2): 401–420. https://doi.org/10.1007/s10798-019-09560-2

Alatta, R. A., and A. Freewan. 2017. "Investigating the Effect of Employing Immersive Virtual Environment on Enhancing Spatial Perception within Design Process." *International Journal of Architectural Research: ArchNet-IJAR* 11 (2): 219. https://doi.org/10.26687/archnet-ijar.v11i2.1258

Alymani, A. 2024. "Integrating Artificial Intelligence Rendering Tools in Design Integrating AI as teaching methods in architectural education." *Proceedings of the 42nd Conference on Education and Research in Computer Aided Architectural Design in Europe*, 629–638.

Autodesk. 2025, September 4. What is Autodesk Forma?

Balbo, R. 2024. "Hybrid Architecture: A Studio-Based Pedagogical Framework for Digital Architectural Design." *Journal of Civil Engineering and Architecture* 18 (8): 363–372.

Botchway, E. A., S. Amos-Abanyie, and S. O. Afram. 2015. "The Impact of Computer Aided Architectural Design Tools on Architectural Design Education. The Case of KNUST." *Journal of Architectural Engineering Technology* 4:145.

Caetano, I., and A. Leitão. 2020. "Architecture Meets Computation: An Overview of the Evolution of Computational Design Approaches in Architecture." *Architectural Science Review* 63 (2): 165–174. https://doi.org/10.1080/00038628.2019.1680524

Calixto, V., and J. Croffi. 2024. "Back to Black Boxes?: An Urgent Call for Discussing the Impacts of the Emergent AI-Driven Tools in the Architecture Design Education." *International Conference on Computer-Aided Architectural Design Research in Asia, CAADRIA* 2024:39–48.

Carpo, M. 2023. *Beyond Digital: Design and Automation at the end of Modernity*. Cambridge: MIT Press.

Carpo, M. 2024. "Generative AI, Imitation Learning, and the Automation of Tacit Knowledge." *A+ u: Architecture and Urbanism* 646:104–109.

Celani, G. C. 2008. "Teaching CAD Programming to Architecture Students." *Gestão & Tecnologia de Projetos* 3 (2): 1–23. https://doi.org/10.4237/gtp.v3i2.73

Chayaamor-Heil, N., V. Perricone, P. Gruber, and F. Guéna. 2023. "Bioinspired, Biobased and Living Material Designs: A Review of Recent Research in Architecture and Construction." *Bioinspiration & Biomimetics* 18 (4): 041001. https://doi.org/10.1088/1748-3190/acd82e

Chester, I.. 2008. "3D-CAD: Modern Technology-Outdated Pedagogy? Exploring Technology Education: Solutions to Issues in a Globalised World." In *Proc., of the 5th Biennial International Conference on Technology Education: Volume One (TERC 2008) Volume One (TERC 2008)*, 75–83.

Dawoud, H. M., H. Al-Samarraie, and F. Zaqout. 2015. "The Role of Flow Experience and CAD Tools in Facilitating Creative Behaviours for Architecture Design Students." *International Journal of Technology and Design Education* 25 (4): 541–561. https://doi.org/10.1007/s10798-014-9294-8

Del Campo, M., and N. Leach. 2022. "Can Machines Hallucinate Architecture? AI as Design Method." *Architectural Design* 92 (3): 6–13.

Doyle, S., L. Forehand, and N. Senske. 2017. "Computational Feminism: Searching for Cyborgs." In *Acadia 2017: Disciplines and Disruption – Proceedings of the 37th Annual Conference of the Association for Computer Aided Design in Architecture*, edited by Takehiko Nagakura, 232–237. Bar Harbor, ME: ACADIA Publishing Company. https://doi.org/10.52842/conf.acadia.2017.232.

Doyle, S., and N. Senske. 2017a. "Between Design and Digital: Bridging the Gaps in Architectural Education." *Charrette* 4 (1): 101–116.

Doyle, S., and N. Senske. 2017b. "Democratizing Access and Identifying Inequalities: Gender, Technology, Architecture." In *ARCC 2017: Architecture of Complexity*, 56–62.

Duarte, J. P., G. Celani, and R. Pupo. 2012. "Inserting Computational Technologies in Architectural Curricula." In *Computational Design Methods and Technologies: Applications in CAD, CAM and CAE Education*, edited by Gu Ning and Xiangyu Wang, 390–411. Hershey, PA: IGI Global.

El Moussaoui, M., and K. Krois. 2023. "Architectural Pedagogy in the Age of AI: The Transformation of a Domain." In *School of Architecture(s) - New Frontiers of Architectural Education. EAAE AC 2023. Springer Series in Design and Innovation*. Vol. 47, edited by M. Barosio, E. Vigliocco, and S. Gomes , 87–93. Cham: Springer. https://doi.org/10.1007/978-3-031-71959-2_11.

Geyer, S. 2024. *Teaching Architecture (s) in the Post-Covid Era: The New Age of Digital Design*. Abingdon: Taylor & Francis.

Iranmanesh, A., and Z. Onur. 2021. "Mandatory Virtual Design Studio for all: Exploring the Transformations of Architectural Education Amidst the Global Pandemic." *International Journal of Art and Design Education* 40 (1): 251–267. https://doi.org/10.1111/jade.12350.

Jenny, D., H. Mayer, P. Aejmelaeus-Lindström, F. Gramazio, and M. Kohler. 2022. "A Pedagogy of Digital Materiality: Integrated Design and Robotic Fabrication Projects of the Master of Advanced Studies in Architecture and Digital Fabrication." *Architecture, Structures and Construction* 2 (4): 649–660. https://doi.org/10.1007/s44150-022-00040-1

J Hopfe, C., V. Soebarto, D. Crawley, and R. Rawal. 2017. "Understanding the Differences of Integrating Building Performance Simulation in the Architectural Education System." *Building Simulation* 15:1249–1256.

Kara, L. 2015. "A Critical Look at the Digital Technologies in Architectural Education: When, Where, and How?" *Procedia - Social and Behavioral Sciences* 176:526–530. https://doi.org/10.1016/j.sbspro.2015.01.506

Kavakoglu, A. A., B. Almaç, B. Eser, and S. Alaçam. 2022. AI Driven Creativity in Early Design Education.

Leitão, A., F. Cabecinhas, and S. Martins. 2010. "Revisiting the Architecture Curriculum." In *Future Cities, 28th ECAADe Conference Proceedings*, 81–88. Zurich, Switzerland: ETH Zurich.

Mariotti, J., and C. Niblock. 2023. "A Critical Reflection on the Impact of Virtual Design Studio on Curriculum Development and Studio Culture in First-Year Architecture Studies." *Trends in Higher Education* 2 (4): 599–610. https://doi.org/10.3390/higheredu2040036

Mark, E., B. Martens, and R. Oxman. 2001. "The Ideal Computer Curriculum." In *Architectural Information Management: Proceedings of the 19th Conference on Education and Research in Computer Aided Architectural Design in Europe (eCAADe 2001)*, 168–175. Helsinki, Finland: Helsinki University of Technology.

Mark, E., B. Martens, and R. Oxman. 2003. "Preliminary Stages of CAAD Education." *Automation in Construction* 12 (6): 661–670. https://doi.org/10.1016/S0926-5805(03)00045-1

Marx, J. 2000. "A Proposal for Alternative Methods for Teaching Digital Design." *Automation in Construction* 9 (1): 19–35. https://doi.org/10.1016/S0926-5805(99)00049-7

Mikhailov, S., A. Mikhailova, N. Nadyrshine, and L. Nadyrshine. 2020. "BIM-technologies and Digital Modeling in Educational Architectural Design." *IOP Conference Series: Materials Science and Engineering* 890 (1): 012168. https://doi.org/10.1088/1757-899X/890/1/012168

Mitchell, W. J. 1975. "The Theoretical Foundation of Computer-Aided Architectural Design." *Environment and Planning b: Planning and Design* 2 (2): 127–150. https://doi.org/10.1068/b020127

Okeil, A. 2010. "Hybrid Design Environments: Immersive and Non-Immersive Architectural Design." *Journal of Information Technology in Construction (ITcon)* 15 (16): 202–216.

Oprean, D., D. Verniz, J. Zhao, J. O. Wallgrün, T. Baird, J. P. Duarte, and A. Klippel. 2021. "Conceptualizing the Remote Site Experience through Immersive Technology: Unraveling the Santa Marta Favela from Students' Perspectives." *Landscape Journal* 39 (2): 31–49. https://doi.org/10.3368/lj.39.2.31

Oprean, D., D. Verniz, J. Zhao, J. O. Wallgrün, J. P. Duarte, and A. Klippel. 2018. "Remote Studio Site Experiences: Investigating the Potential to Develop the Immersive Site Visit." In *23rd International Conference on Computer-Aided Architectural Design Research in Asia: Learning, Prototyping and Adapting, CAADRIA 2018*, 421–430.

Ostrowska-Wawryniuk, K., M. Strzała, and J. Słyk. 2022. "Form Follows Parameter: Algorithmic-Thinking-Oriented Course for Early-Stage Architectural Education." *Nexus Network Journal* 24 (2): 503–522. https://doi.org/10.1007/s00004-022-00603-1

Oxman, R. 2008. "Digital Architecture as a Challenge for Design Pedagogy: Theory, Knowledge, Models and Medium." *Design Studies* 29 (2): 99–120. https://doi.org/10.1016/j.destud.2007.12.003

Oxman, R., and R. Oxman. 2010. "The New Structuralism: Design, Engineering and Architectural Technologies." *Architectural Design* 80 (4): 14–23. https://doi.org/10.1002/ad.1101

Özgen, D. S., Y. Afacan, and E. Sürer. 2021. "Usability of Virtual Reality for Basic Design Education: A Comparative Study with Paper-Based Design." *International Journal of Technology and Design Education* 31 (2): 357–377. https://doi.org/10.1007/s10798-019-09554-0

Ozkar, M. 2017. *Rethinking Basic Design in Architectural Education: Foundations Past and Future*. New York: Routledge.

Pelsmakers, S., E. Donovan, J. Poutanen, A. Hoggard, N. Sandström, and K. Friedman. 2021. "Architecture Design Studio in Uncertain Times: Digital Teaching and Learning Implications." *Scroope* 30:47–72.

Petric, J., and T. Maver. 2003. "Digital Prototyping in the Architectural Design Studio." In *SIGraDi 2003-[Proceedings of the 7th Iberoamerican Congress of Digital Graphics] Rosario Argentina,* , 5–7.

PlanFinder. 2025. Draw Plans Faster. https://www.planfinder.xyz/.

Pressman, A. 2012. *Designing Architecture: The Elements of Process*. London: Routledge. https://doi.org/10.4324/9780203122174.

Robertson, B. F., J. Walther, and D. F. Radcliffe. 2007. Creativity and the Use of CAD Tools: Lessons for Engineering Design Education from Industry.

Salleh, R., M. A. M. Yusoff, S. C. Amat, A. M. Noor, and N. Suredah. 2013. "Profiling Industry-Relevant Competencies of Graduate Architect through Online Job Advertisements." *International Business Research* 6 (11): 43. https://doi.org/10.5539/ibr.v6n11p43

Schnabel, M. A. 2006. "Architectural Parametric Designing." In *Communicating Space(s): Proceedings of the 24th Conference on Education and Research in Computer Aided Architectural Design in Europe (eCAADe 2006)*, 216–221. Volos, Greece: University of Thessaly.

Solemma. 2025, September 4. Climate Studio.

Soliman, S., D. Taha, and Z. El Sayad. 2019. "Architectural Education in the Digital age: Computer Applications: Between Academia and Practice." *Alexandria Engineering Journal* 58 (2): 809–818. https://doi.org/10.1016/j.aej.2019.05.016

The Architects' Council of Europe. 2025. The Architectural Profession in Europe 2024 Sector Study.

Uysal, V. Ş., and F. Topaloğlu. 2017. "Bridging the Gap: A Manual Primer into Design Computing in the Context of Basic Design Education." *International Journal of Art & Design Education* 36 (1): 21–38. https://doi.org/10.1111/jade.12048

Vazquez, E. 2024. "Teaching Parametric Design: Fostering Algorithmic Thinking through Incomplete Recipes." *Open House International* 49 (4): 736–751. https://doi.org/10.1108/OHI-06-2023-0135